



Master's Thesis

Data Consistency Evaluation for Reaction Model Optimization

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Abstract

This Master’s thesis presents the development of a computational framework for the uncertainty quantification and consistency analysis of chemical kinetic models. The work focuses on bridging the gap between raw experimental data and the systematic optimization of reaction mechanisms through the implementation of four core modules designed for the TUMKIN data infrastructure. First, an experimental shock tube IDT uncertainty estimation package was developed. This module quantifies uncertainty by combining random (Type-A) and systematic (Type-B) components. Systematic errors are evaluated through facility specific factors, allowing for a standardized assessment of non-idealities in shock tube measurements. Second, a reaction rate coefficient (RRC) uncertainty estimation package was implemented to programmatically assess kinetic parameter reliability. By collecting data across multiple libraries in the RMG database, the package evaluates rate coefficients over a standardized temperature grid. It utilizes a reparameterized log-linear Arrhenius model and Nonlinear Least Squares to derive temperature-dependent uncertainty factors (f) for each reaction. Third, the framework features robust Cantera IDT simulations to support the generation of training data. This module is capable of handling diverse experimental ignition definitions (e.g., OH max, dP/dt max, baseline intercept) to ensure high-fidelity comparisons between model and experiment. These simulations provide the foundation for sensitivity analysis, where impact factors are used to identify active reactions. Finally, these components were integrated with the Bound-to-Bound Data Collaboration (B2BDC) framework. By building quadratic surrogate models from the simulation results, the framework calculates a Scalar Consistency Measure (SCM) to evaluate the agreement between experimental datasets and the reaction mechanism. The function of the integrated pipeline is demonstrated through consistency checks on a joint Syngas/Hydrogen mechanism, identifying inconsistent data points and defining feasible parameter spaces. This work establishes a rigorous, automated path for data-driven kinetic model optimization.

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1 Introduction

1.1 Motivation

The development of accurate chemical kinetic mechanisms is fundamental to the design and optimization of sustainable energy systems and advanced aerospace propulsion. Modern combustion research has transitioned from manually validated models toward automated frameworks

Current efforts, such as the TUMKIN infrastructure, aim to revive and expand upon foundational data collaboration concepts like PrIme. The objective is to create a comprehensive environment that facilitates the analysis, optimization, and uncertainty quantification of reaction mechanisms across a wide range of fuels.

1.2 Problem Statement

A primary challenge in chemical kinetics is the inherent uncertainty in both experimental data and model parameters. Reaction rate coefficients (RRCs) often vary across literature sources due to different fitting techniques or theoretical calculations, leading to significant spreads in predicted combustion properties. Simultaneously, experimental measurements—such as shock tube ignition delay times (IDT)—are subject to systematic and random uncertainties that must be rigorously quantified to be useful for model validation. If a reaction model fails to accurately predict experimental results within established uncertainty intervals, it indicates a fundamental inconsistency. This inconsistency may arise from "bad" experimental data, such as gross errors or outliers. Traditional validation methods often struggle to distinguish between these two sources of error, particularly when dealing with large-scale datasets and multivariate parameter spaces.

1.3 Objectives

The primary objective of this thesis is to implement a systematic workflow for evaluating data consistency in the context of reaction model optimization. This involves the integration of experimental data with predictive simulations to identify and rectify inconsistencies. The specific goals are as follows:

Develop a methodology to calculate uncertainty factors for reaction rate coefficients by performing nonlinear least squares fits on data. Implement and test the methodology on kinetics databases.

Implement a method to quantify the systematic and random uncertainties of shock tube ignition delay time measurements.

Perform consistency analysis using the Bound-to-Bound Data Collaboration (B2BDC) framework to establish a feasible parameter region where model predictions and experimental observations overlap within their deterministic bounds.

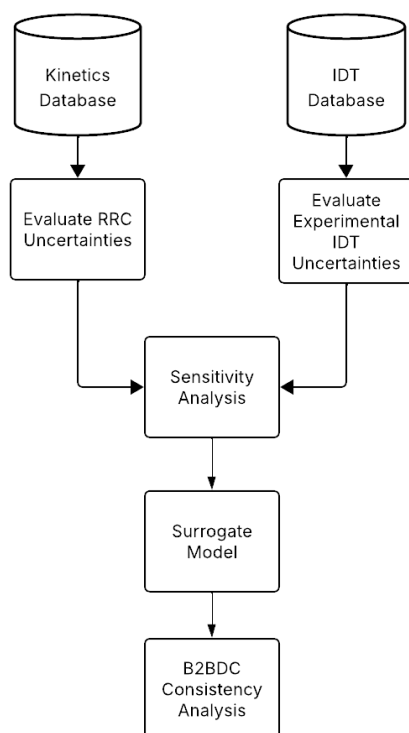


Figure 1.1: Flow diagram overview of the workflow of this thesis.

2 Background and State of the Art

2.1 Chemical Kinetics Databases

TUMKIN, a chemical kinetic data infrastructure currently under development aims to incorporate databases and tools to facilitate analysis and optimization of reaction mechanisms of wide range of fuels. The idea is based on Process Informatics Model (PrIMe) created by Frenklach and coworkers which was a data infrastructure concentrating on data management, data uncertainty, and data consistency. [1] However PrIMe website is not active. TUMKIN aims to revive and expand on PrIMe. A similar effort is RESPECTH [2]. Their database contains direct and indirect experimental measurements for various fuels. Relevant for this thesis are the indirect shock tube ignition delay time measurements for Ethylene, Syngas and Hydrogen. [3][4][5] They also share computer programs for various tasks around analysis and optimization of kinetics. Reaction Mechanism Generator (RMG) is an automatic chemical reaction mechanism generator that constructs kinetic models composed of elementary chemical reaction steps using a general understanding of how molecules react. RMG database contain kinetics information for many reactions from various sources and datasets. [6]

Pelucchi et al. [7] describe a shift in combustion chemistry from handmade validation to the development of "methods able to automatically validate the generated models". They emphasize the need for Automation. The many-model problem, where numerous mechanisms representing the same fuels use different rate constants and pathways creates an increased need of comparisons with experimental data to improve the reliability and performances of the models" They propose a "quantitative, comprehensive approach to estimate the agreement between models and experimental data" that accounts for the "distance between measurements and predictions," "shapes of the interpolating curves," and "possible shifts on the independent variable". The authors emphasize that "effective algorithms and codes to compare models and measurements are required" alongside standardized data formats (such as RKD or YAML) to ensure "consistency, accountability, traceability, and accuracy".

SciExpeM [8], is a "collection of services to automatically collect, manage, and analyze experimental data and models". Within the "numerical combustion domain," this system supports a "continuous improvement methodology" composed of three separate steps:

- 1.) "after the predictive model is developed, using the proper numerical tools, it is used to simulate the available experimental datasets";
- 2.) "the output obtained from the numerical simulations is compared to the experimental datasets, and a quantitative answer about the accuracy of such predictions is provided";
- 3.) "according to the results of the validation phase, the model is updated to improve the predictions in the weak performing areas, and the loop is restarted to check again the model performance against the whole experimental database".

To facilitate this validation, the framework implements "the Curve Matching similarity measure," which "provides an index in the $[0,1]$ range to evaluate how close the results of a simulation are to the experimental data, taking into account also the uncertainty of experimental data". This methodology is based on "functional analysis... of both the experimental data points and the model predictions, performed through a spline based interpolation of both data series". The agreement is quantified through the "evaluation of distance indices, based on L^2 norm, and similarity indices, based on the Pearson Correlation, for both functions and the related derivatives".

2.2 Uncertainty Quantification in Combustion

Uncertainty quantification (UQ) is the science for quantification and minimization of uncertainties with computer experiments and simulation. [9] The approach to developing detailed kinetic models (mechanisms) of fuel combustion involves compiling a set of elementary reactions whose rate parameters may be determined from individual rate measurements, reaction rate theory, or a combination of both. These methods have uncertainties. A widely used method is sensitivity analysis. UQ is closely related to sensitivity analysis, and in many ways, UQ is a natural extension of sensitivity analysis. A basic objective of UQ is to analyze the uncertainty in the model response because of model parameters being uncertain, including forward uncertainty quantification and propagation and an inverse problem leading to model uncertainty constraining. When treated as a Bayesian inference problem, UQ aids the development of predictive kinetic models by constraining models with data and yielding estimates for the confidence of a model to make predictions outside of the thermodynamic regimes where the model has been tested. [9]

2.3 Uncertainty of Reaction Mechanisms

Reaction mechanisms employed in chemical kinetics models include information about the relevant species, reactions, reaction rates and the corresponding uncertainties of these reaction rates. Reactions are modeled with the Arrhenius equation with k being the reaction rate coefficient (RRC). In the literature, Arrhenius coefficients (A, n, E_a) are fitted to experimental data with a regression model and evaluated with expert opinion. Or they are calculated from fundamental calculations. Therefore, different literature sources can report different reaction rate coefficient $k(T)$ values for a given reaction. The uncertainty factor f is a logarithmic measure of the plausible spread of $k(T)$ around its recommended value:

$$k(T) = AT^n \exp\left(-\frac{E_a}{T}\right) \quad (\text{cm}^3 \text{ mol}^{-1} \text{ s}^{-1}, \text{ K}) \quad (2.1)$$

To perform analysis or optimization of combustion mechanisms, it is important to quantify the uncertainty of its parameters. For every reaction there are many literature sources reporting varying parameter sets. The most comprehensive source of a database compiling such sources is NIST Kinetics Database. Parameter standard deviations $s(\log_{10}(A))$, $s(n)$, $s(E_a)$ are used to calculate lower and upper uncertainty bounds for rate coefficients.

$$k_{\text{low}}(T) = 10^{\log_{10}(A) - s(\log_{10} A)} T^{n - s(n)} \exp\left(-\frac{E_a + s(E_a)}{T}\right) \quad (2.2)$$

$$k_{\text{up}}(T) = 10^{\log_{10}(A) + s(\log_{10} A)} T^{n + s(n)} \exp\left(-\frac{E_a - s(E_a)}{T}\right) \quad (2.3)$$

The uncertainty parameter f is defined as:

$$f(T) = \log_{10}\left(\frac{k_{\text{up}}(T)}{k_0(T)}\right) = \log_{10}\left(\frac{k_0(T)}{k_{\text{low}}(T)}\right) \quad (2.4)$$

Where k_0 is the best fit of the rate coefficient, k_{low} and k_{up} are the lower and the upper bounds respectively [10]. However, authors rarely report covariance matrix or parameter standard deviations. When the covariance matrix is not available, often only pre-exponential factor A is varied to achieve the prescribed uncertainty band on $k(T)$ [11].

2.4 Sensitivity Analysis

Local sensitivity analysis is a method where the response of the model to small parameter changes close to nominal values is explored. Normalized first-order sensitivity coefficients are

commonly applied in combustion to provide kinetic insight into the model and its strengths and weaknesses with respect to experimental data.[12]

Tomlin et al. [13] employ global uncertainty analysis for three mechanisms describing the low-temperature oxidation of dimethyl ether with application to simulations of species concentrations corresponding to existing data from an atmospheric pressure flow reactor and high-pressure ignition delays. The study aimed to evaluate how inherent uncertainties in their input data may impact on the predictive capability of the schemes with respect to available target experimental data. Assessing the sensitivity of selected predictive targets to potential uncertainties within the input data can assist in identifying the most important reactions within a scheme, which determine the accuracy to which key combustion properties can be predicted.

2.5 Surrogate Models

In the B2BDC framework, surrogate models are constructed using a solution-mapping approach over the prior uncertainty region. Active parameters are first identified through impact-factor screening and the surrogate is built only in this reduced parameter space. Training data are generated using space-filling designs, and the original model is evaluated at each design point. Polynomial, quadratic, or rational quadratic surrogate models are then fitted by minimizing either L_2 or L_∞ norms. Rational quadratic surrogates are used when polynomial models are insufficiently accurate, and their coefficients are obtained through semidefinite programming [14].

2.6 Consistency Analysis

Given a reaction mechanism, chemical kinetics simulation software like Cantera[15] can simulate combustion, and predict properties like ignition delay time(IDT), laminar flame speed(LFS), etc. These properties are also measured in real life combustion experiments, and reported together with quantified uncertainties. A good reaction model should be able to predict these properties accurately under same operating conditions as the physical experiment(temperature, pressure, stoichiometric ratio, mixture content). Consistency analysis in this context are methods to validate simulation and experimental results against each other by finding a set of model parameters that can explain the experiment data within their uncertainty intervals. If some data can't be explained by any model, this suggests data and

the model are inconsistent. The inconsistency could be caused by the shortcomings of the model or the data.

2.6.1 Anomaly Detection / Gross Error Detection

Ridzuan and Zainon [19]”Data cleansing offers a better data quality which will be a great help for the organization to make sure their data is ready for the analyzing phase. This paper reviews the data cleansing process, the challenge of data cleansing for big data and the available data cleansing methods. Data cleansing process mainly consists of identifying the errors, detecting the errors and correcting them.”

Rousseeuw and Hubert [21], Real world datasets often contain anomalous cases, also known as outliers. A useful tool for this purpose is robust statistics, which aims to detect the outliers by first fitting the majority of the data and then flagging data points that deviate from it. To avoid the masking and swamping effects of classical methods, the goal of robust statistics is to find a fit which is close to the fit we would have found without the outliers. Outliers are then identified by their large ‘deviation’ (e.g., its distance or residual) from that robust fit. In the multivariate case, a robust measure of location $\hat{\mu}$ and scatter $\hat{\Sigma}$ can be used to compute robust distances for any point x :

$$RD(x, \hat{\mu}, \hat{\Sigma}) = \sqrt{(x - \hat{\mu})^T \hat{\Sigma}^{-1} (x - \hat{\mu})} \quad (2.5)$$

Robust estimates can be obtained via the Minimum Covariance Determinant (MCD) method, which looks for those h observations in the dataset whose classical covariance matrix has the lowest possible determinant. Alternatively, in a linear regression setting, the Least Trimmed Squares (LTS) estimator identifies outliers by minimizing the sum of the h smallest squared residuals:

$$\text{minimize } \sum_{i=1}^h (r^2)(i) \quad (2.6)$$

where $(r^2)(1) < (r^2)(2) < \dots < (r^2)(n)$ are the ordered squared residuals.

Chandola et al. [22]. Anomaly detection refers to the problem of finding patterns in data that do not conform to expected behavior. The importance of anomaly detection is due to the fact that anomalies in data translate to significant (and often critical) actionable information in a wide variety of application domains. A straightforward anomaly detection approach, therefore, is to define a region representing normal behavior and declare any observation in the data which does not belong to this normal region as an anomaly.” For consistency analysis

involving statistical comparisons between observed and expected values, the survey highlights the use of the χ^2 statistic. "The value of the χ^2 statistic is determined as:

$$\chi^2 = \sum_{i=1}^n \frac{(X_i - E_i)^2}{E_i} \quad (2.7)$$

where X_i is the observed value of the i_{th} variable, E_i is the expected value of the i_{th} variable (obtained from the training data) and n is the number of variables."

Tong and Crowe[23]. "The theory of data reconciliation has been developed to resolve the contradictions between the measurements and their constraints and to process contaminated data into consistent information". They present a "new class of test statistics for detection and identification of gross errors based on principal component analysis". Principal component analysis (PCA) is used to transform "a set of correlated variables into a new set of uncorrelated variables, known as principal components (PCs)". For a linear steady state process, the residuals of reduced constraints are defined as:

$$e = B\tilde{x} \quad (2.8)$$

where $\tilde{x}$ is a vector of the measured variables and B is the balance matrix. The principal component transform of the residual vector is given by:

$$y_e = W_e^T(e - e^*) = W_e^T e \quad (2.9)$$

where y_e consists of principal components and W_e forms eigenvectors of the covariance matrix H_e . It is shown that this "new test is capable of detecting gross errors of small magnitudes and has substantial power to correctly identify the variables in error, when the other tests fail".

2.6.2 Bayesian Approaches

Bell et al.[24] First principles Markov Chain Monte Carlo sampling is used to investigate uncertainty quantification and uncertainty propagation in parameters describing hydrogen kinetics. Their goal is to characterize the distribution of parameters that are consistent with a given set of experimental data. The resulting distribution identifies strong positive and negative correlations and several non-Gaussian characteristics. These results indicate the degree to which parameters consistent with the target experiments constrain predicted behavior in different combustion regimes.

Braman et al.[25] use a Bayesian framework to investigate the capabilities of syngas combustion models through the quantification of uncertainties. This framework, given a set of experimental data, "allows for the calibration of model parameters, determination of uncertainty in those parameters, propagation of that uncertainty into simulations, as well as determination of model evidence from a set of candidate syngas models". The state of knowledge about a parameter value is represented by a probability density function (PDF), which is updated according to Bayes' theorem:

$$P_{post}(\theta|d) = \frac{P_{prior}(\theta)\pi(\theta; d)}{\int P_{prior}(\theta)\pi(\theta; d)d\theta} \quad (2.10)$$

where P_{prior} is the prior PDF, P_{post} is the posterior PDF, and $\pi(\theta; d)$ is the likelihood function. The likelihood function "quantifies the level of agreement between the model and the data for specific values of the parameters". For model comparison, the authors employ the evidence function, which "measures the consistency of the model and the data considering the entire parameter space":

$$\pi_{evid}(M_i; d) = \int P_{prior}(\theta_i|M_i)\pi(\theta_i, M_i; d)d\theta_i \quad (2.11)$$

The study evaluates this approach with a "particular focus on kinetics parameter calibration and evidence based chemistry model comparison" using laminar flame speed measurements from high pressure experiments.

Yousefian et al.[26] proposes a novel toolchain is for Bayesian inference and uncertainty quantification (UQ) studies in practical hydrogen enriched and lean premixed combustion systems. The methodology implements an integrated chemical reactor network (CRN), nonintrusive polynomial chaos expansion using the point collocation method (NIPCE-PCM), and the Markov Chain Monte Carlo (MCMC) method. The stochastic process for quantities of interest (QoI), $\hat{y}$, is represented using a polynomial chaos expansion (PCE) defined as:

$$\hat{y} = \sum_{i=1}^p \alpha_i \Psi_i(\theta) \quad (2.12)$$

where θ is the vector of uncertain input parameters and $\Psi_i(\theta)$ represents multivariate polynomials. To reconcile experiments with the model, Bayes' theorem is implemented to transform a prior probability distribution, $P(\theta)$, for unknown parameters into a posterior distribution, $P(\theta|y)$:

$$P(\theta|y) = \frac{\mathcal{L}(\theta)P(\theta)}{\int \mathcal{L}(\theta)P(\theta)d\theta} \quad (2.13)$$

where $P(y)$ is experimental data and $\mathcal{L}(\theta) = P(y|\theta)$ represents a likelihood function.

Sheen and Wang [27] use MUM-PCE, where the uncertainty in the rate parameters of a model is first quantified using stochastic spectral expansion and polynomial chaos techniques. The model is then constrained against global combustion experiments through solution mapping and least squares optimization, combined with a consistency screening procedure to identify incompatible experimental targets. The uncertainty of the constrained (posterior) model is subsequently calculated using an inverse spectral technique and propagated to a range of combustion conditions. [27]

2.6.3 Bound-to-Bound Data Collaboration (B2BDC)

Bound-to-Bound Data Collaboration (B2BDC) provides a natural framework for addressing both forward and inverse uncertainty quantification problems [28]. In this approach, quantity of interest (QOI) models are constrained by related experimental observations with interval uncertainty [28]. QOI can be for example ignition delay time, laminar flame speed or species concentrations. Uncertainties in model parameters and experimental data are characterized by deterministic bounds and propagated to bounds of prediction uncertainty, resonating “Bound-to-Bound” in the name [14].

A collection of models and observations is termed a dataset and carves out a feasible region in the parameter space [28]. The framework considers a set of n uncertain model parameters, denoted as $x = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^n$ [14]. Prior knowledge about these parameters is typically expressed as independent interval bounds:

$$l_i \leq x_i \leq u_i, \quad i = 1, \dots, n \quad (2.14)$$

[14] which defines a hypercube $\mathcal{H}$ in the parameter space [14]. In the context of this thesis parameters are reaction rate coefficients. And their interval bounds will be calculated by statistical analysis of various kinetics data sources.

Experimental observations are represented by models $M_e(x)$ and their corresponding experimental uncertainty bounds $[L_e, U_e]$ [14]. In this thesis, the experimental observation of interest is the ignition delay time. Uncertainty bounds on experimental ignition delay times are estimated by the method that will be described in Section 3.1. The requirement that model predictions must lie within experimental uncertainty generates inequality constraints on model parameters:

$$L_e \leq M_e(x) \leq U_e, \quad e \in \mathcal{E} \quad (2.15)$$

[14] where $\mathcal{E}$ is the set of experimental indices [14]. It is therefore important to assign model parameter intervals and experiment uncertainty bounds carefully. When experimental

uncertainty bounds or model parameter intervals are too loose, the model would always be able to explain the data, so consistency analysis loses its meaning. The collection of all such constraints defines a region in the model parameter space termed the feasible set $\mathcal{F}$:

$$\mathcal{F} = \{x \in \mathcal{H} \mid L_e \leq M_e(x) \leq U_e, \forall e \in \mathcal{E}\} \quad (2.16)$$

The agreement/disagreement among models and data is determined by inspecting whether there exist parameter vectors in the feasible set: The dataset is consistent if the feasible set is not empty and inconsistent otherwise [14]. Numerically, examination of dataset consistency is accomplished by calculating a quantity termed the Scalar Consistency Measure (SCM), $C_{D,S}$. In the implementation of [14] SCM is calculated as an interval by two different optimization methods. When inner and outer bound solutions are both positive, SCM also has to be positive, therefore a feasible set exists. Conversely if inner and outer bound solutions are both negative, SCM is negative and no feasible set of parameters exist. If the inner bound solution is negative and outer bound solution positive, in that case SCM is unable to conclude consistency.

The SCM is defined as the solution to the following constrained optimization problem:

$$C_{D,S} = \max_{x \in \mathcal{H}, \gamma \in \mathbb{R}} \gamma \quad (2.17)$$

$$\text{subject to } L_e + \gamma\delta_e \leq M_e(x) \leq U_e - \gamma\delta_e, \quad \forall e \in \mathcal{E} \quad (2.18)$$

[14] where $\delta_e = \frac{U_e - L_e}{2}$ represents the half width of the experimental uncertainty interval [14]. The dataset is inconsistent if the scalar consistency measure is negative ($C_{D,S} < 0$) and consistent otherwise ($C_{D,S} \geq 0$) [14].

Prediction uncertainty for an unmeasured QOI, $M_p(x)$, is computed by finding the minimum and maximum values of the prediction model within the feasible set [14]:

$$L_{pred} = \min_{x \in \mathcal{F}} M_p(x) \quad (2.19)$$

$$U_{pred} = \max_{x \in \mathcal{F}} M_p(x) \quad (2.20)$$

[14]. The interval $[L_{pred}, U_{pred}]$ represents the ‘‘collaborative’’ prediction, as it utilizes the combined information from the prior parameter bounds and the entire experimental dataset to constrain the possible values of the QOI [14], [29].

In real world applications, it is often the case that collections of models and observations are inconsistent [28]. To identify which models and/or observations are problematic, the

Vector Consistency Measure (VCM) strategy aims to recover dataset consistency by relaxing the fewest experimental uncertainty bounds [14]. The VCM approach introduces a vector of relaxation coefficients to investigate datasets with numerous sources of inconsistency [28]. VCM can make predictions on which parameter intervals or experimental bounds need to be relaxed in order to achieve consistency. It can also be useful in the case where SCM is inconclusive. VCM can suggest relaxations that achieve consistency.

The constrained optimization problems involved in B2BDC are generally nonconvex and difficult to solve globally as nonlinear optimization solvers usually converge to a local optimum [14]. To address this, solution mapping techniques are used to create polynomial and rational quadratic surrogate models [14]. With these surrogates, convex relaxation techniques (such as semidefinite programming) are used to derive provable, conservative bounds on the solution of the original nonconvex problems [14], [29]. Another practical benefit of using surrogate models is that they are a lot cheaper to evaluate than Cantera or Chemkin models. A MATLAB implementation of B2BDC is provided by [14].

2.7 Model discrepancy

Model discrepancy is the difference between model behavior and real data that can't be removed by model parameter optimization. In the context of reaction mechanisms it might be that an important reaction is missing in the mechanism. No amount of parameter optimization likely will not capture that missing behavior. Or it might be that model parameters only the reaction rate coefficients due to practical constraints. And that some other fixed parameters are not good enough. The discrepancy caused by the fixed wrong parameters cannot be reconciled with by optimizing the parameters at hand. One possible method of identifying model discrepancy is to check the residuals of the model and the data for structure. If residuals are biased, model doesn't capture enough. In that case adding a model discrepancy term and fitting it as a polynomial might be helpful. [16]

If model discrepancy is due to shortcomings of experiments (i.e. scenario dependent) we would expect it to work best when modeled as different functions for different QOIs. However if it is a shortcoming of the model (i.e. missing reactions) a single function (e.g. polynomial) fit or a single prior distribution of the discrepancy might be enough to partially compensate.

Kennedy and O'Hagan [17] provide a foundational Bayesian framework for the calibration of computer models that explicitly accounts for various sources of uncertainty, most notably model inadequacy.

3 Methodology

The packages implemented in this thesis have been developed considering the needs of the currently under development TUMKIN chemical kinetics data infrastructure. Kinetics database and the IDT database shown in Figure 3.1 currently refer to RMG Database [6], and RESPECTH IDT shock tube database [2]. Once these packages get integrated into TUMKIN, these databases will be replaced with the internal TUMKIN database. The next step of uncertainty estimation for IDT data and for reaction rate coefficients are not dependent on any operating condition or reaction mechanism. They can therefore be run once on the database, and the results stored. Sensitivity Analysis step requires, a reaction mechanism .yaml file and conditions at which sensitivity and consequently consistency analyses will be conducted. Sensitivity analysis results in a set of active reactions. User can specify their own active reactions if they want to. Unless otherwise specified, RRCs and RRC uncertainties will be taken from the evaluated uncertainties if enough sources exist. For reactions that can't be assigned uncertainties this way, RRC from the user supplied mechanism will be used. The user has the option to interject their own RRC and uncertainty factors if they want to. After this step quadratic surrogate models for each operating condition is created by the B2BDC toolbox. Which is subsequently used in consistency analysis. A detailed explanation of the process is given in the following sections. Source code for the packages can be found at: <https://github.com/barkinguney/consistency-analysis>

3.1 Experimental Shock Tube IDT Uncertainty

Every dataset in TUMKIN needs to contain appropriate uncertainties. If published sources don't report uncertainties for experimental data, uncertainties are estimated using the algorithm proposed by [30]

The approach is aligned with the Guide to the Expression of Uncertainty in Measurement (GUM) [31], Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results [32]. The base principle follows from [31] and [32]: "the result of a measurement is only an approximation or estimate of the value of the specific quantity subject to measurement,

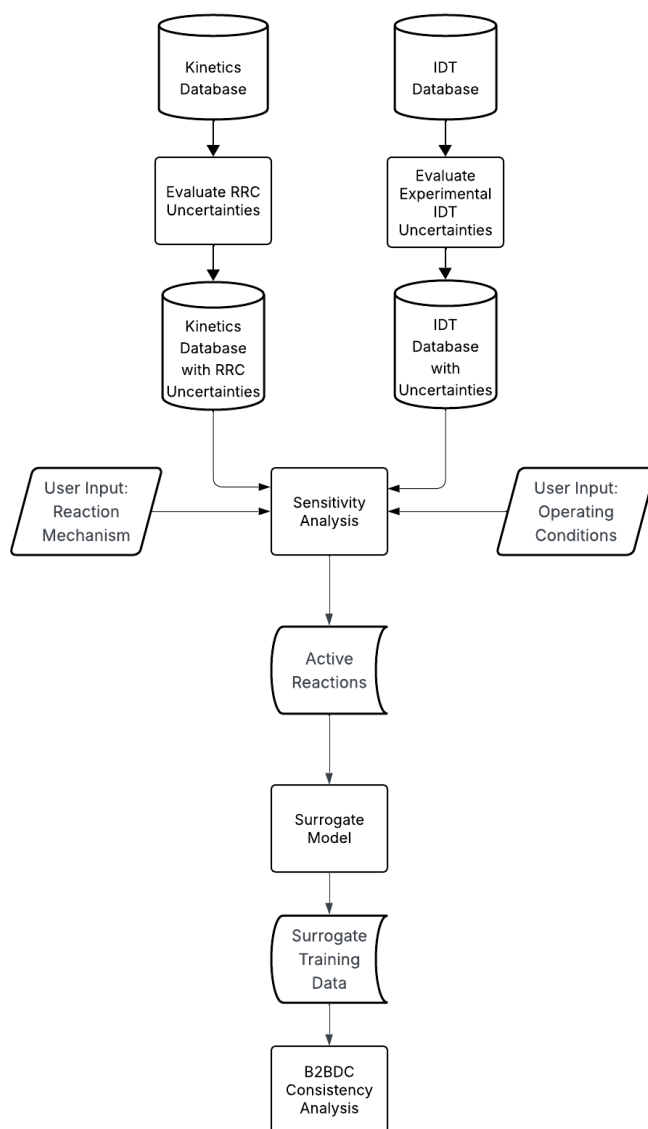


Figure 3.1: Flow diagram showing the steps for consistency analysis.

that is, the measurand, y , and thus the result is complete only when accompanied by a quantitative statement of its uncertainty”.

The reporting uncertainty is the expanded uncertainty U together with the coverage factor k used to obtain it, or report combined uncertainty u_c , which represents the estimated standard deviation of the result.

The reporting a measurement result and its uncertainty, include the information of all components of standard uncertainty, which may be grouped into two categories according to the method used to estimate their numerical values: type A, those which are evaluated by statistical methods, i.e. the corresponding component of uncertainty arising from a random effect; type B, those which are evaluated by means other than the statistical analysis of series of observations. In this thesis type A and type B will be equivalently referred to as uncertainty components “random”, denoted $\sigma_r(y)$, and “systematic,” denoted $\sigma_\delta(y)$ respectively. The combined standard uncertainty is calculated as the square root of the sum of the variances from systematic and random components

$$u_c(y) = \sqrt{\sigma_\delta(y)^2 + \sigma_r(y)^2} \quad (3.1)$$

$\sigma_r(y)^2$ is estimated combining the individual standard uncertainties σ_i of the uncertainty components applying the law of uncertainty propagation.

The *systematic error* $\sigma_\delta(y)$ captures device- and condition-specific biases in τ_{ign} measurements in shock tubes. As these conditions are typically modelled using a 0-D mathematical framework that accounts solely for chemical processes behind of shock wave, i.e. an ideal case. Alternative IDT modeling approaches are outside the present scope. To evaluate $\sigma_\delta(y)$, a statistical procedure that quantifies the departure from this idealization is developed. Concretely: (i) the dominant facility/operating-condition factors provoke non-idealities behind of shock wave are identified; (ii) construct estimators for each factor’s contribution to $\sigma_\delta(y)$; (iii) define a quasi-ideal reference regime—a subset of facilities/conditions that meet strict criteria—and assign to this reference a baseline systematic error of 5%; (v) establish the quantitative metric of deviation from reference case; and (iv) for factors outside the reference, incremental penalties and aggregate these contributions to obtain $\sigma_\delta(y)$ are computed. This construction makes explicit the assumption that a small, irreducible systematic component ($\approx 5\%$) exists even under best-case conditions, with additional bias driven by quantifiable departures from the ideal case. Consequently, the systematic error can be classified as an “offset or zero setting error”.

Because rigorous, quantitative assessments of systematic error in τ_{ign} data remain scarce, the primary focus of this work is on addressing this gap; details follow in the next sections.

The final uncertainty is expressed as a coverage interval, where U is the expanded uncertainty obtained by multiplying $u_c(y)$ by a coverage factor, k . Thus

$$U = ku_c(y) \quad (3.2)$$

and it is confidently believed that $y - U \leq Y \leq y + U$, which is commonly written as $Y = y \pm U$. To be consistent with current international practice, the value of k to be used for calculating U is, by convention, $k = 2$.

3.1.1 Systematic Uncertainty

Guided by non-ideality phenomena in shock tubes, 14 input/operating factors (IFs) are selected, Figure 3.2, that determine proximity to the 0-D “ideal case”, i.e., conditions under which τ_{ign} can be described with the 0-D mathematical model. A quasi-ideal reference is defined by fixing each factor to its target metric μ_i^0 Figure 3.2 and assign a baseline systematic error $\delta^0 = 5\%$ at 95% confidence. This baseline represents the irreducible risk under best-case conditions; deviations from μ_i^0 are treated separately.

Category	#	Impacting Factor (IF)	Target metric, μ_i^0
Test equipment	1	Driver length	≥ 3.0 m
	2	Driven length	≥ 8.0 m
	3	Driven inner diameter	≥ 10.0 cm
	4	Tailoring	yes
	5	Inserts	yes
	6	CRV	yes
Operating conditions	7	Reflected-shock Temperature, T_5	1000 - 1600 K
	8	Reflected-shock Pressure, P_5	5 - 20 atm
	9	Valid test time, $\Delta t_{\text{valid}}^*$	50 - 500 μs
	10	Dilution (Ar/He/...)**	yes
	11	Impurities	no
	12	Measurement location	$\leq 2\text{cm}$ (from the end of the tube)
	13	Fuel-oxidizer equivalence ratio	≥ 0.3
	14	Pre-shock pressures, P_1	≥ 10 Torr
*Δt_{valid} time window before contact-surface/expansion-wave arrival or significant boundary-layer growth invalidates 0-D assumptions. ** “Yes” indicates purposeful dilution to suppress non-ideal gas dynamics and heat-release effects (typical practice is high inert fraction; specify per system if needed).			

Figure 3.2: Impacting factors (IF) and target metrics for the “ideal case” $\delta_0 = 5\%$ (95% confidence).

GUM-consistent recipe to turn Type-B systematic term δ into a combined uncertainty $u_c(y)$, Eq.3.1, with Type-A random. So, calculated in this section δ tis systematic expanded relative uncertainty will be later transformed in the systematic standard uncertainty.

The systematic error is calibrated against a quasi-ideal reference through departures of IFs from its reference target. Assuming the independence of all other IFs these departures are mapped to incremental penalties via a smooth “undesirability” transform, and aggregated as

$$\delta = \delta^0 + \sum_{i=1}^N \delta_i = \delta^0 (1 + \sum_{i=1}^N \xi_i), \quad \delta_i = \delta^0 \xi_i, \quad i = 1, \dots, N, \quad (3.3)$$

where N is the number of IFs, and $\xi_i \in [0, 1)$ quantifies the penalty induced by the deviation of μ_i from the ideal μ_i^0 .

To define and ξ_i , a two-sided, smaller-is-better modification of Harrington’s desirability concept [33] [34] retaining low sensitivity near the ideal point and saturation for large deviations is used:

$$\xi_i = 1 - e^{-(x_i^2)} \quad (3.4)$$

$$x_i = (\mu_i^0 - \mu_i) / \mu_i^0. \quad (3.5)$$

For binary IFs (e.g., tailoring, inserts, CRV, dilution, impurities), an unfavorable setting is mapped to the conservative lower edge of the “satisfactory” band i.e. $\xi_i = e^{-1} \cong 0.37$, if “no” and to $\xi_i = 0$ if “yes” for IF_{4,5,6,10,13}. For IF₁₁ it is otherwise.

As the test time (IF₉) is strongly correlated with T₅ and P₅, IF₉ is retained in the systematic term and, T₅ and P₅ excluded from δ (their uncertainties enter the random part later). IF₁₄ (P₁) is also excluded due to a lack of quantitative evidence.

3.1.2 Random Uncertainty

The measured value should be reported along with an estimate of the total combined standard uncertainty, sum of systematic error and random error. Formally, the rigorous approach to evaluate the random error of a measurement would require the repetition of the same measurement for a significant number of times, what is non-realistic to perform in praxis or from the rules for differentiating the measurement model describing the indirect measured target. In this case, the measured data set is approximated with the function of several

parameters and the standard deviation of a measurand is calculated as the positive square root of the combined variance, $u_c^2(y)$, assuming independence of input parameters [31]:

$$u_c^2(y) = \sum_{i=1}^N \left(\frac{\partial f}{\partial z_i} \right)^2 u^2(z_i) + 2 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{\partial f}{\partial z_i} \frac{\partial f}{\partial z_j} u(z_i) u(z_j) r(z_i, z_j), \quad (3.6)$$

where z_i are parameters of the function f ; $u^2(z_i)$ is the standard uncertainty of parameters; $r(z_i, z_j)$ is the estimated correlation coefficient.

To describe the set of measured IDT data the function proposed by A. Lifshitz and co-authors for data processing is used[35]:

$$\tau_{ign} = f(z_i, A, B, C, D) = A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D. \quad (3.7)$$

Here $z_1 = T_5$ is the temperature behind the shock wave, $z_2 = P_5$ is the pressure behind the shock wave, and $z_3 = \varphi$ stoichiometric ratio, fuel/oxygen. A, B, C, D – parameters of approximation.

Eq. 3.6 and Eq. 3.7 are applied under assumptions A, B, C, D, T_5, P_5 , and φ are not correlated parameters and terms with parameter correlations can be neglected.

In this case

$$\begin{aligned} \sigma_i^2(\tau_{ign}) = & \left(z_2^C z_3^D A \exp\left(\frac{B}{z_1}\right) \cdot \left(-\frac{B}{z_1^2}\right) \right)^2 \cdot u^2(z_1) + \left(C A \exp\left(\frac{B}{z_1}\right) z_2^{C-1} z_3^D \right)^2 \cdot u^2(z_2) \\ & + \left(D A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^{D-1} \right)^2 \cdot u^2(z_3) + \left(\exp\left(\frac{B}{z_1}\right) z_2^C z_3^D \right)^2 \cdot u^2(A) \\ & + \left(\frac{1}{z_1} A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D \right)^2 \cdot u^2(B) + \left(A \exp\left(\frac{B}{z_1}\right) z_2^C \ln(z_2) z_3^D \right)^2 \cdot u^2(C) \\ & + \left(A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D \ln(z_3) \right)^2 \cdot u^2(D) \end{aligned} \quad (3.8)$$

The standard uncertainty of input parameters $u^2(z_i)$, i.e. manufacture's specifications, calibration, and other certificates should be reported in a paper. If it is not so it can be evaluated from the available information, analogy, etc.

Weighted Least Squares (WLS) regression is used to calculate best A, B, C, D parameters and their uncertainties $u^2(A), u^2(B), u^2(C), u^2(D)$. If all data points in a data set have very similar values for any of the variables T_5, P_5, ϕ the fit is multicollinear. This makes the calculated uncertainties meaningless. This is especially common for ϕ since an IDT dataset usually contains measurements at constant ϕ . To prevent this, if a variable has constant value

across the dataset, Equation 3.7 is modified to set the corresponding parameter B,C or D set to 1. In the case of constant ϕ , Equation 3.7 becomes:

$$\tau_{ign} = f(z_i, A, B, C) = A \exp\left(\frac{B}{z_1}\right) z_2^C z_3. \quad (3.9)$$

Equation 3.8, that propagates parameter uncertainties to τ_{ign} uncertainty is also recalculated with the new parameter set A,B,C.

Normally, T_5 , P_5 , and ϕ are reported in a paper with uncertainty interval given as

$$Z = z \pm U(z). \quad (3.10)$$

Assuming normal distribution of x_i and 95% of confidence interval one can interpret U_i as uncertainty interval

$$U(z_i) = 2\sqrt{u_c^2(z_i)} \quad (3.11)$$

and the standard uncertainty of parameters can be calculated as

$$u^2(z_i) = \frac{U^2(z_i)}{4}. \quad (3.12)$$

The standard uncertainty of a set of IDT data can be calculated from the sum of systematic and random uncertainties

$$u_{ci}(\tau_{ign}) = \sqrt{\sigma_{\delta_i}^2 + \sigma_i^2(\tau_{ign})} \quad (3.13)$$

$\sigma_{\delta_i}^2$ is the standard systematic deviation. As δ is interpreted as normal, 95% expanded, results in:

$$\sigma_{\delta_i}^2 = \left[\tau_{ign} * \left(\frac{\delta_i/2}{100} \right) \right]^2 \quad (3.14)$$

The final uncertainty interval of the measured set will be calculated as (in the case of normal distribution of x_i and 95% of confidence interval):

$$Ui(\tau_{ign}) = 2(u_{ci}(\tau_{ign})) \quad (3.15)$$

Finally, data accepted for the further analysis and model testing, fitting and optimization are reported as:

$$Y^{exp} = \tau_{ign}^m \pm U(\tau_{ign}) \quad (3.16)$$

where τ_{ign}^m is measured value of the IDT.

In addition to the facility specific information needed for systematic error calculation, measurement uncertainties for T_5 , P_5 and ϕ also need to be reported by papers for accurate

uncertainty estimation. If they are not reported, a default values of $\pm 1\%$ is used for each in this thesis.

3.2 Arrhenius Parameters Uncertainty

Different literature sources report different values for the Arrhenius parameters or rate coefficients for the same reaction. Therefore, it is important to take all this information into account when analysing and optimizing reaction mechanisms. This package collects kinetic data from a database and assigns a single uncertainty factor f to each reaction, using the collected data.

3.2.1 Data Collection

To evaluate the uncertainty of a specific reaction, all instances of that reaction across all loaded kinetic libraries in the RMG (Reaction Mechanism Generator) database are identified. Reactant and product species are determined by comparing the reaction equation string to known species. SMILES representations of the species are used to search RMG database libraries for matching sources. For every match, rate coefficients $k(T)$ are evaluated across a grid of the valid temperature range of each reaction. A downside of RMG database is very few sources include a valid temperature range for the reported kinetic parameters. In that case a standardized temperature grid $T \in [300, 3100]$ K is used to collect rate coefficient data. Collected kinetics data, and rate coefficients evaluated at fixed intervals within the temperature range are saved further processing. Currently RMG kinetics types of Arrhenius, MultiArrhenius, ThirdBody are supported. For future development, support for PDepArrhenius should be added. Currently efficiencies of ThirdBody reactions are not considered.

3.2.2 Uncertainty Factor

The modified Arrhenius equation is expressed as:

$$k(T) = AT^n \exp\left(-\frac{E_a}{RT}\right) \quad (3.17)$$

To perform a robust statistical fit across multiple orders of magnitude, equation is transformed into logarithmic space.

$$\ln k(T) = \ln A + n \ln T - \frac{E_a}{RT} \quad (3.18)$$

In standard Arrhenius fitting parameters (A, n, E_a) are usually highly correlated.

$$\ln k(T) = \beta_0 + n \ln \left(\frac{T}{T_{ref}} \right) - \frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \quad (3.19)$$

The rate expression is transformed into a log linear model centered around a reference temperature, which is geometric mean of temperature unless otherwise specified. The parameters (β_0, n, E_a) are solved using a Nonlinear Least Squares approach, minimizing the residual sum of squares (RSS) between the library data and the model. The resulting covariance matrix in the reparameterized space, Σ_β , is then mapped back to the original $(\log_{10} A, n, E_a)$ space using the Jacobian transformation matrix J :

$$\Sigma_\theta = J \Sigma_\beta J^T \quad (3.20)$$

Diagonal entries of the covariance matrix of the least squares fit give parameter standard deviations $s(\log_{10}(A))$, $s(n)$, $s(E_a)$, which are used to calculate lower and upper uncertainty bounds for rate coefficients.

$$k_{low}(T) = 10^{\log_{10}(A) - s(\log_{10} A)} T^{n - s(n)} \exp \left(-\frac{E_a + s(E_a)}{T} \right) \quad (3.21)$$

$$k_{up}(T) = 10^{\log_{10}(A) + s(\log_{10} A)} T^{n + s(n)} \exp \left(-\frac{E_a - s(E_a)}{T} \right) \quad (3.22)$$

The uncertainty parameter f is defined as:

$$f(T) = \log_{10} \left(\frac{k_{up}(T)}{k_0(T)} \right) = \log_{10} \left(\frac{k_0(T)}{k_{low}(T)} \right) \quad (3.23)$$

Where k_0 is the best fit of the rate coefficient, k_{low} and k_{up} are the lower and the upper bounds respectively [10]. This gives a temperature dependent uncertainty factor $f(T)$. Temperature dependence of $f(T)$ is usually low. To simplify further analysis, maximum value of $f(T)$ is selected as constant f . When rate coefficients need to be sampled in further analysis, they are sampled from the interval $[k_0(T)/10^f(T), k_0(T) * 10^f(T)]$. In practice this can be achieved by multiplying the pre exponential factor A with a multiplier sampled from $[(T)/10^f(T), (T) * 10^f(T)]$. Which preserves the shape of the $k(T)$ curve and only scales it.

$$f_{final} = \max_T f(T) \quad (3.24)$$

This uncertainty factor provides a single, conservative metric to quantify uncertainty of the reaction rate coefficient across the entire temperature range.

3.3 Cantera IDT Simulations

Cantera is a 0D and 1D chemical kinetics simulation software. To simulate shock tube IDT, constant volume batch reactors are used. Ignition delay times to be simulated for a given operating conditions T5, P5 phi and mechanism. The exact measurement or simulation time of IDT vary based on IDT definition. Using the same IDT definition as reported by the physical experiment to calculate simulated IDT would yield the closest results to physical experiment. The ignition types that can be handled by this IDT simulation code are as follow. Following the format described by RESPECTH shock tube IDT measurements, ignition type is comprised of attributes, target, type, amount and unit . Target can be pressure, temperature or concentration. Type can be “max”, “d/dt max”, “baseline max/min intercept from d/dt”, “concentration”, “relative concentration”, and “relative increase”. “max” means IDT is at the maximum of the target property. “d/dt max” means IDT is at the maximum gradient of target property. “baseline max/min intercept from d/dt” means the time when the max d/dt line intercepts the baseline value of target property. “concentration”: IDT is at the time target property reaches a specified concentration value. “relative concentration”: IDT is at the time when the concentration of the target property relative to its maximum concentration reach the specified value. “relative increase”: IDT is at the time when the value of the target property increases by a given relative amount compared to its value at $t=0$. Amount attribute is only used when type are “concentration”, “relative concentration” or “relative increase”. Unit attribute is only used when the type is “concentration”, “relative concentration” or “relative increase”. Unit should have the value “unitless” for relative types.

3.4 Consistency analysis

Consistency methods, that check for a given dataset of experimental ignition delay time measurements and a mechanism with parameter bounds, whether experiments and mechanism are consistent with each other. B2BDC is used as the consistency analysis method in this thesis.

3.4.1 Sensitivity analysis

For a given QoI, the model responses are sensitive to only a small subset of parameters. These parameters, referred to as active parameters, dominate the response behavior, a phenomenon referred to as effect sparsity [36]. Employing a sensitivity analysis method to identify these active parameters and using only these active parameters in further analysis improves computational feasibility. In [14] active parameters are identified by sampling the prior uncertainty region using space-filling designs like Latin hypercube sampling [37] or Sobol quasi-random sequence [38], and evaluating the original model at those points. A linear regression model S_l is then fitted to approximate the QoI as a function of the parameters.

$$S_l = \sum_{i=1}^n c_i x_i + c_0 \quad (3.25)$$

S_l contains information about how much the model output changes when each parameter is varied around its nominal value. The resulting regression coefficients define impact factors, which estimate how much the model output changes when each parameter varies across its uncertainty range. This is similar to sensitivity, but it also considers how large the parameter uncertainties are. A parameter with low sensitivity can still be high impact if it has high uncertainty.

$$IF_i = |c_i| (u_i - l_i) = |c_i| \Delta x_i \quad (3.26)$$

Active parameters are selected by screening the ranked impact factors with a user set threshold [14]. If the IF of a reaction at an operating condition is low, that reaction is considered "low impact". If a reaction is low impact across all operating conditions, it is then flagged "inactive". All "active" reactions are considered for further analysis. It is possible to specify whether IF_i or $|c_i|$ should be used to select active parameters. Using IF_i would include information about the size of parameter uncertainty in determining active reactions.

3.4.2 Polynomial Surrogate Modeling for B2BDC

3.4.3 B2BDC Integration

3.5 Data Sources

When TUMKIN is up and running, uncertainty quantification and consistency analysis codes described in this thesis will use the data available in TUMKIN. The following sources has been used in the mean time to develop these modules, and could be imported into

TUMKIN if suitable. Kinetics database RESPECTH [2] collect IDT, laminar flame speed, and concentration measurements for various fuels. They are reported in their xml format. For testing the modules developed in this thesis I have used the RESPECTH data for shock tube IDT data for ethylene, Syngas and Hydrogen. Currently Most comprehensive collection of reaction rates and Arrhenius parameters is the NIST Kinetics Database [39]. That contain many literature sources for every reaction. However, they don't allow access to their database from code, no API or anything. It is the most comprehensive source, however, must be accessed manually. Reaction Mechanism Generator (RMG) on the other hand also contains a kinetics database that is openly accessible. For this thesis Arrhenius parameter data is taken from RMG database. To algorithmically extract kinetics data from RMG for a given reaction, SMILES or INCHI representations of the products and reactants are necessary. So, a species database that map between the widely accepted SMILES, INCHI representations and Cantera species name, and composition is necessary. An example format can be seen in OPTIMA++. Species names for the same species can be different across mechanism files if they are consistent within the mechanism. This is however not suitable for TUMKIN database. So before ingesting Mechanism files into TUMKIN it should be validated that species in mechanism can unambiguously be found in the TUMKIN species data base. This validating step is not in the scope of this thesis however is necessary for TUMKIN.

4 Results and Discussion

In this thesis tools for Arrhenius parameter uncertainty, Experimental IDT uncertainty, sensitivity analysis have been developed and integrated with B2BDC for consistency analysis.

4.1 B2BDC Consistency

Joint Syngas/Hydrogen mechanism from [40] is used. Wang et. al report that among others, experimental IDT data by TAMU[41] show good data consistency with their model. [40]. To test the integration of the B2BDC MATLAB toolbox with this thesis work, same IDT data will be used. Figure 4.3, Figure 4.4, Figure 4.6, Figure 4.7 are generated by the MATLAB toolbox provided by [14]

4.1.1 Consistent Dataset

Cantera IDT simulations are used to generate training data. Quadratic surrogate models for each data point are built from the training data. The B2BDC consistency check results in a scalar consistency measure interval of $[0.6118, 0.7261]$. Since the lower bound is positive, this demonstrates that the dataset and the model are consistent, and a feasible set of model parameters exists.

4.1.2 Inconsistent Dataset

Same dataset as the previous Section is taken. 2 "bad" data points are added to artificially create an inconsistent dataset. Cantera IDT simulations are used to generate training data. Quadratic surrogate models for each data point are built from the training data. The B2BDC consistency check results in a scalar consistency measure interval of $[-3.0078, -0.5989]$. This proves dataset inconsistency, as the conservative bound (outer bound) of the scalar consistency measure is negative [14]. Figure 4.6 and Figure 4.7 rank the observation and

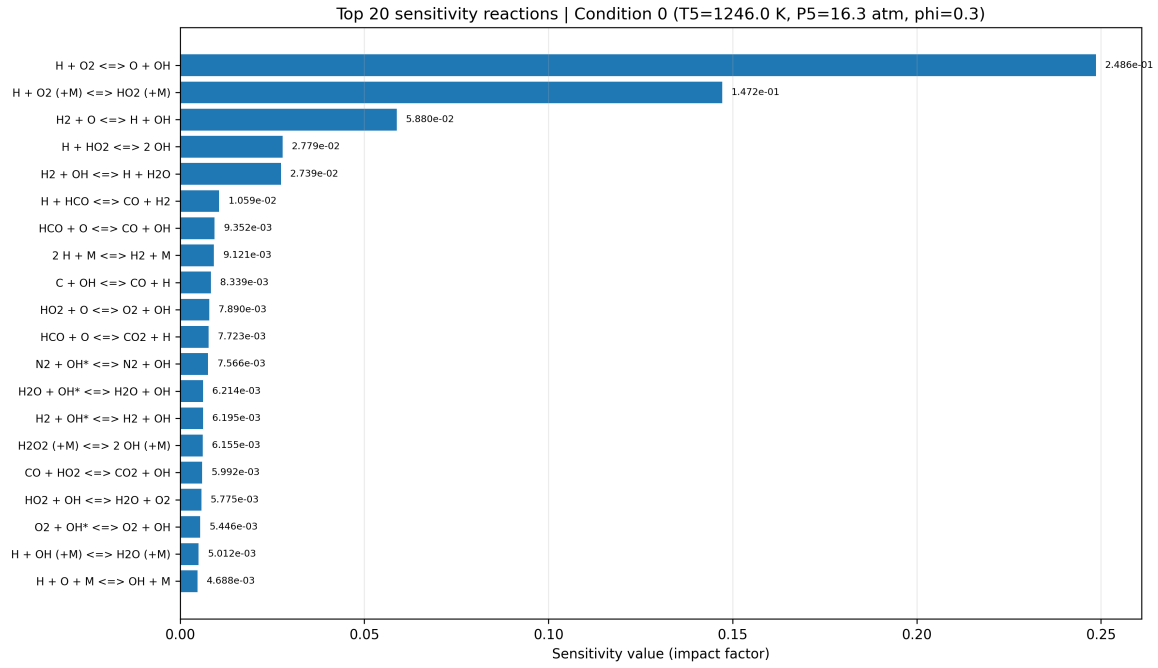


Figure 4.1: Sensitivity of IDT to the reaction rate coefficients at a single operating condition. Condition from [41]. $H_2/O_2/Ar$ mixture, $\phi = 0.3$, $P = 16.3$ atm

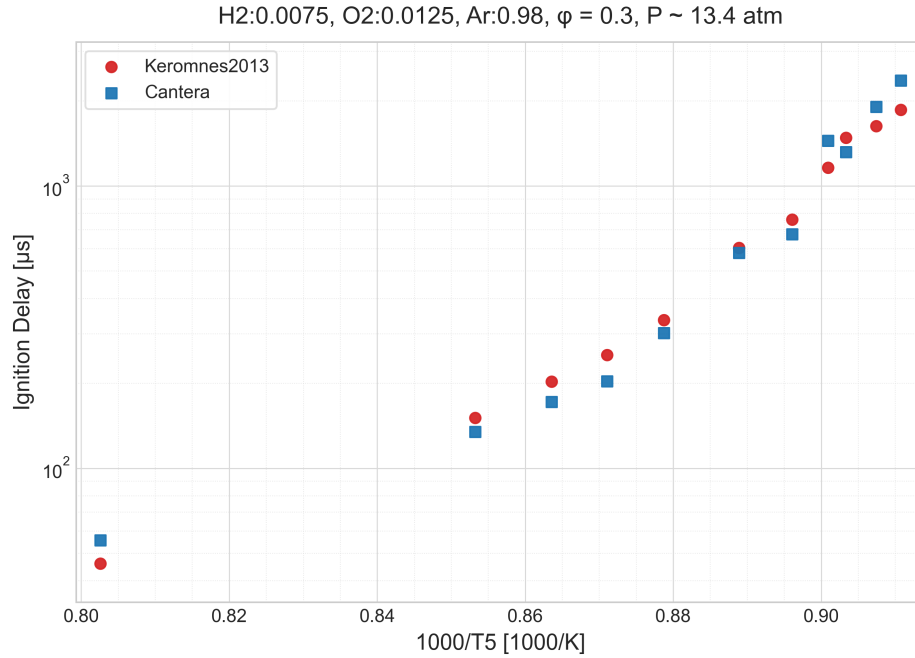


Figure 4.2: IDT measurements [41]. $H_2/O_2/Ar$ mixture, $\phi = 0.3$, $P \sim 13.4$ atm

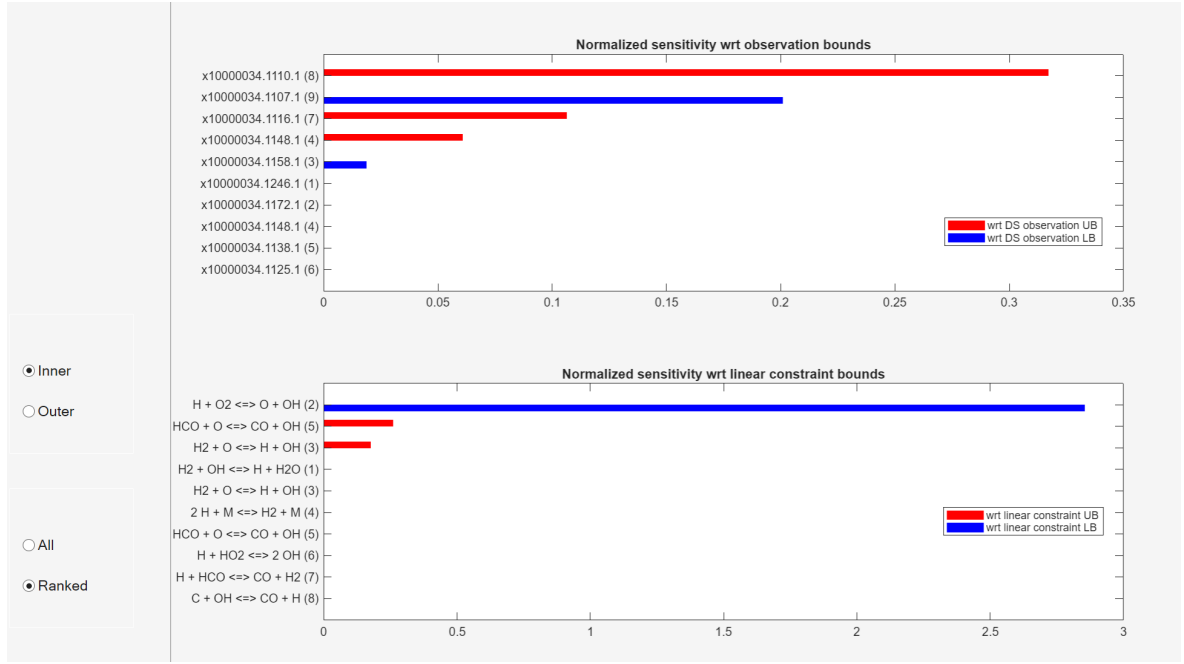


Figure 4.3: Ranked sensitivity of scalar consistency measure (SCM) inner-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds.

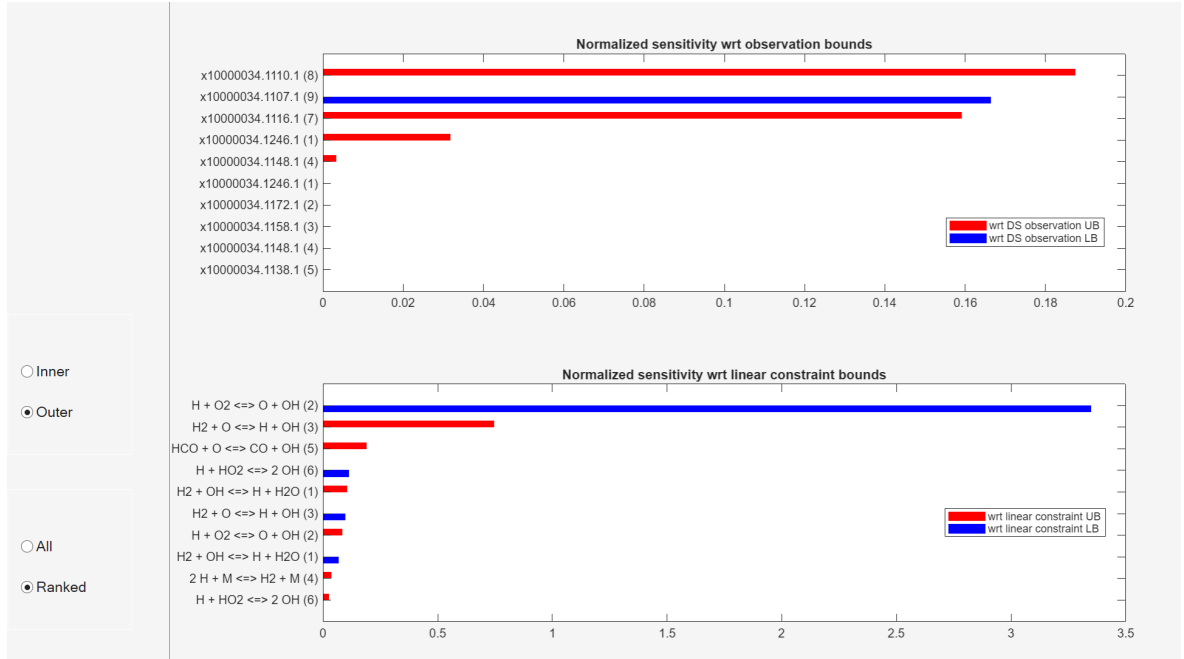


Figure 4.4: Ranked sensitivity of scalar consistency measure (SCM) outer-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds.

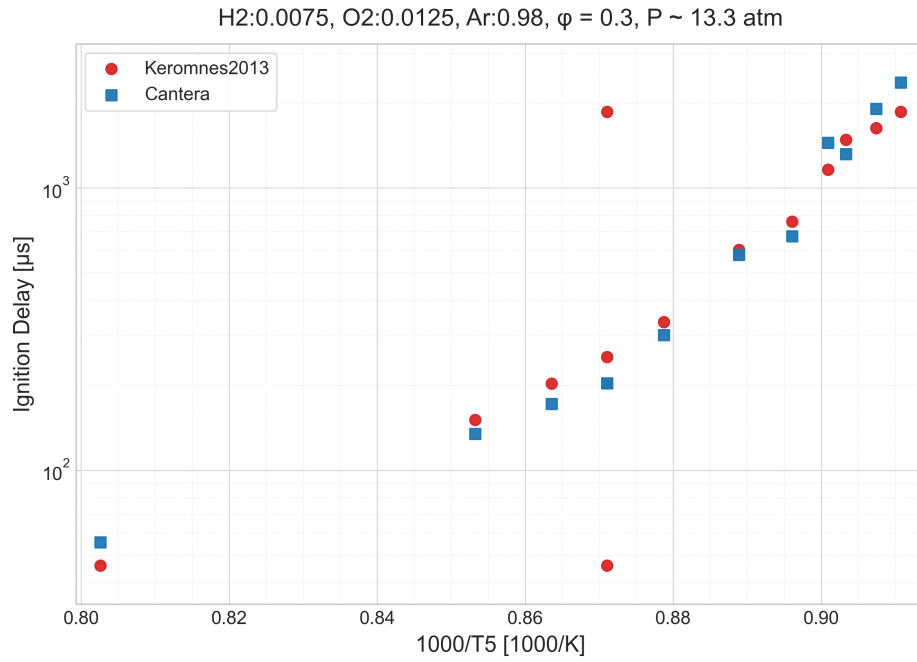


Figure 4.5: IDT measurements [41] + 2 artificial "bad" data points.
 $H_2/O_2/Ar$ mixture, $\phi = 0.3$, $P \sim 13.3$ atm

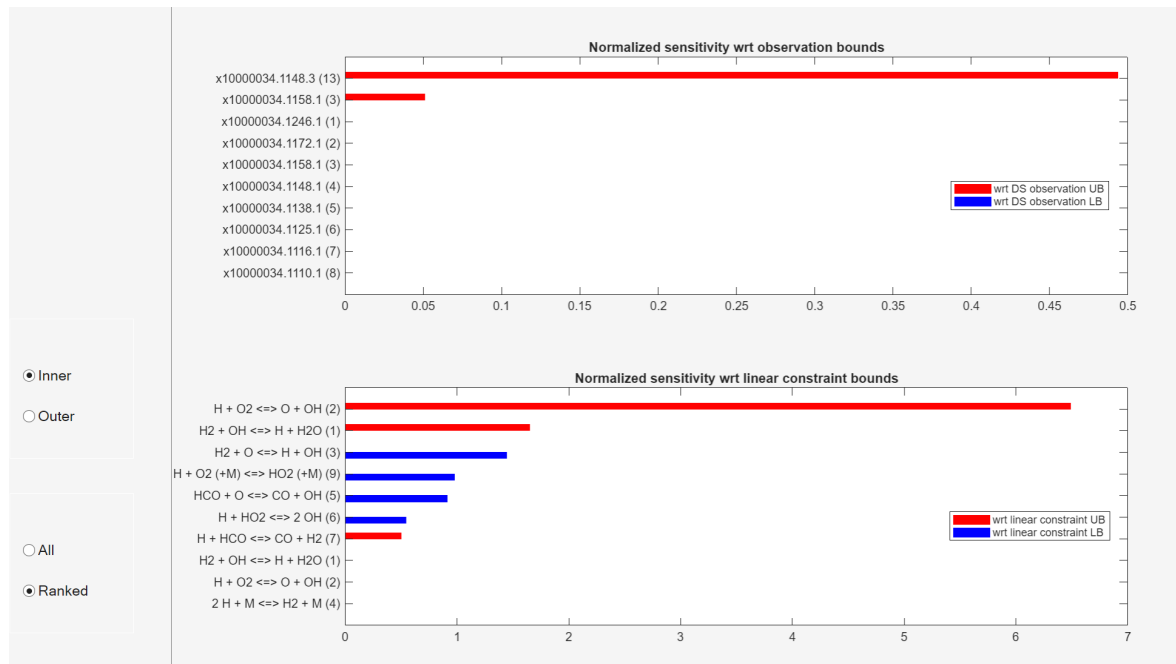


Figure 4.6: Ranked sensitivity of scalar consistency measure (SCM) inner-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds. Dataset includes inconsistent data.

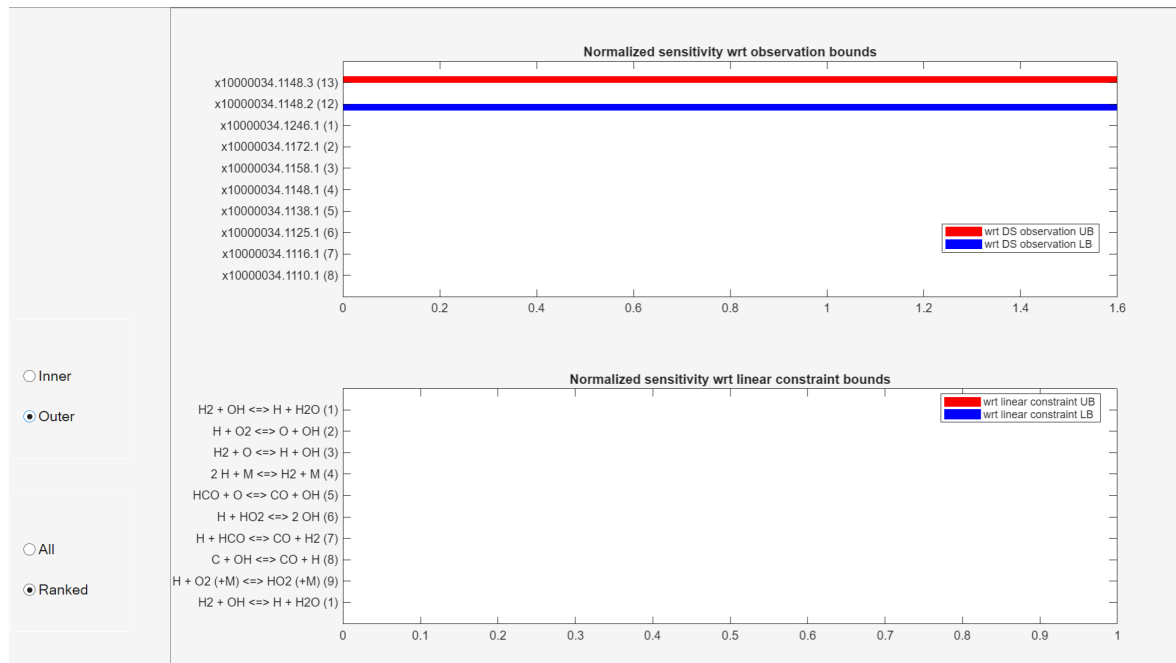


Figure 4.7: Ranked sensitivity of scalar consistency measure (SCM) outer-bound solutions with respect to the quantity of interest (QOI) and model parameter uncertainty bounds. Dataset includes inconsistent data.

constraint bounds to which the consistency is most sensitive. As expected, the artificially added "bad" data points (12) and (13) are the largest suspected causes of the inconsistency. In the next step, these data points are removed from the dataset. The scalar consistency measure now yields $[0.6118, 0.7261]$. Since the lower bound is now positive, this demonstrates that the dataset and the model are consistent, and a feasible set of model parameters exists. Because the dataset and model are now equivalent to those discussed in Section 4.1.1, the sensitivities of the scalar consistency measure's inner and outer bound solutions to the QoI and model parameter uncertainty bounds are also equivalent to Figure 4.3 and Figure 4.4.

4.2 Uncertainty of Shock Tube IDT Data

4.2.1 shen vs davidson

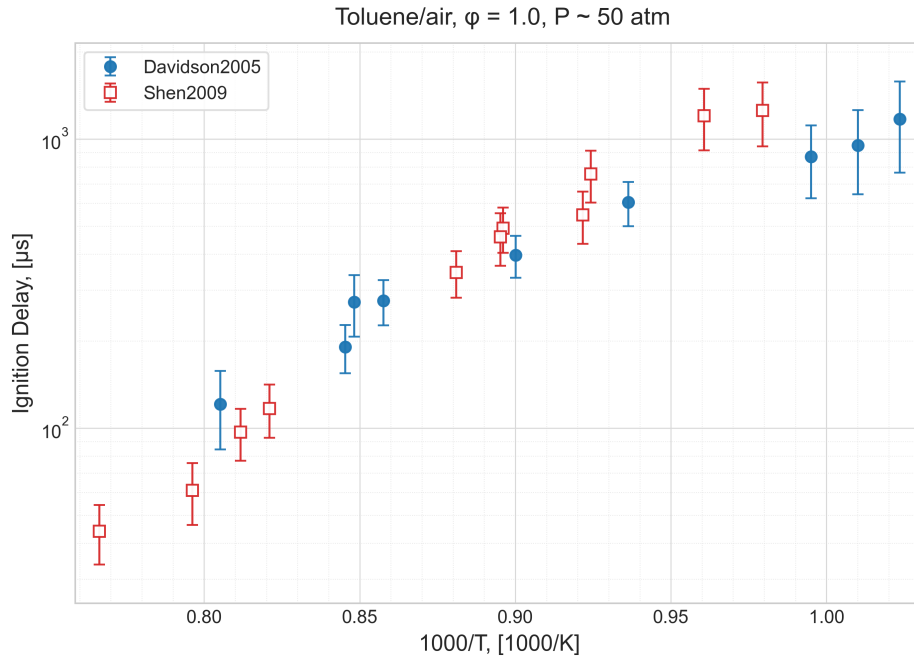


Figure 4.8: Comparison of calculated IDT uncertainties for a stoichiometric toluene/air mixture. IDT measurements are from two different facilities [42][43] over similar temperature range. $P \sim 50$ atm.

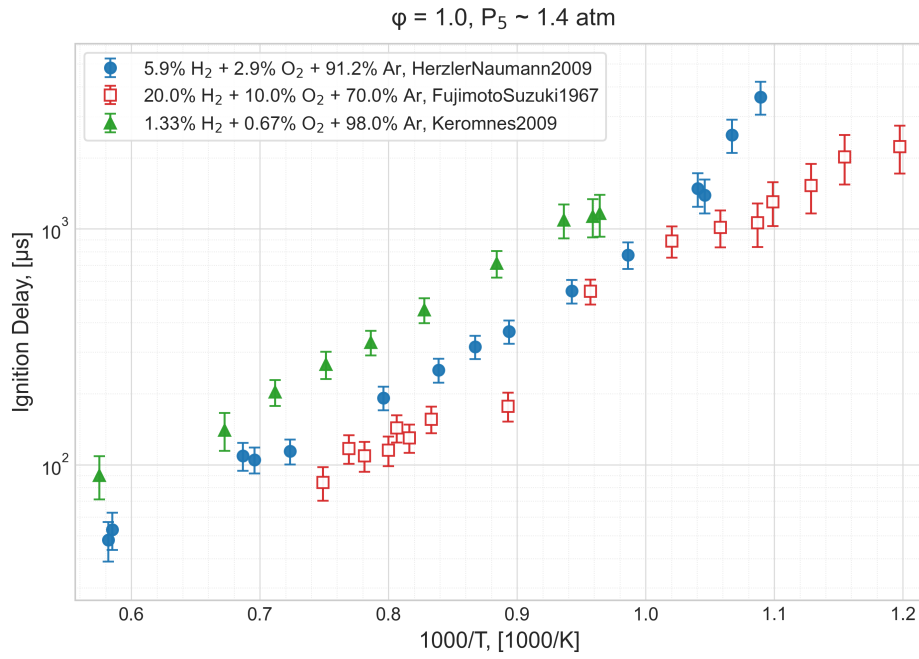


Figure 4.9: Comparison of calculated IDT uncertainties for $\text{H}_2/\text{O}_2/\text{Ar}$ mixtures. IDT measurements are from sources [44] [45] [41]. $\phi = 1.0, P \sim 1.4 \text{ atm}$.

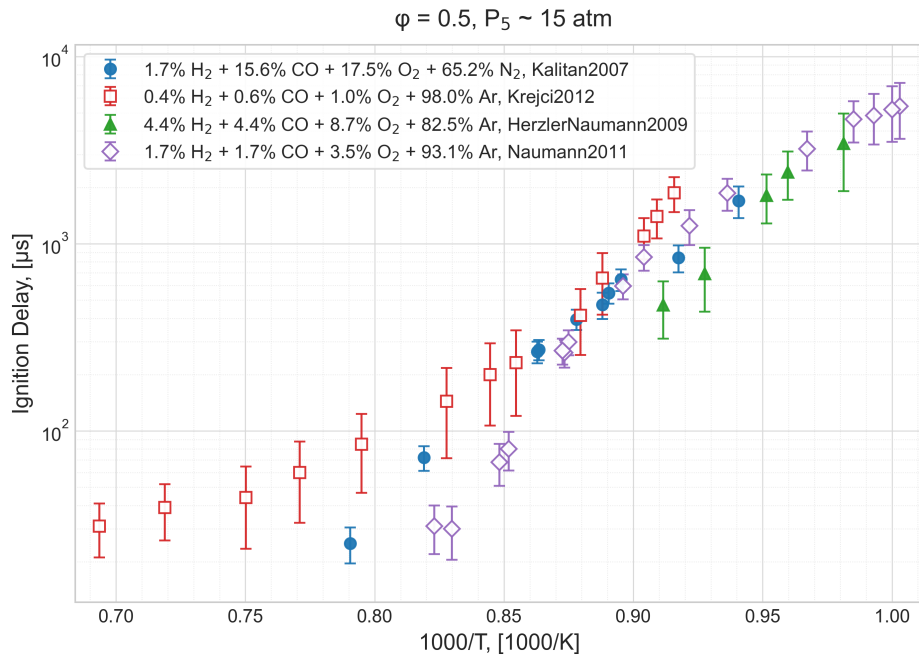


Figure 4.10: Comparison of calculated IDT uncertainties for diluted syngas/oxidizer mixtures. IDT measurements are from sources [46] [47] [44] [48]. $\phi = 0.5, P \sim 15 \text{ atm}$.

4.2.2 Syngas

for most reported sources shock tube setup specific information is not available in an easily accessible database. They are available in the original papers, however to extract them would require manual entry, for programmatic processing with language model. This is necessary for an accurate estimation of experimental shock tube IDT uncertainties.

4.3 Reaction Rate Coefficient Uncertainty

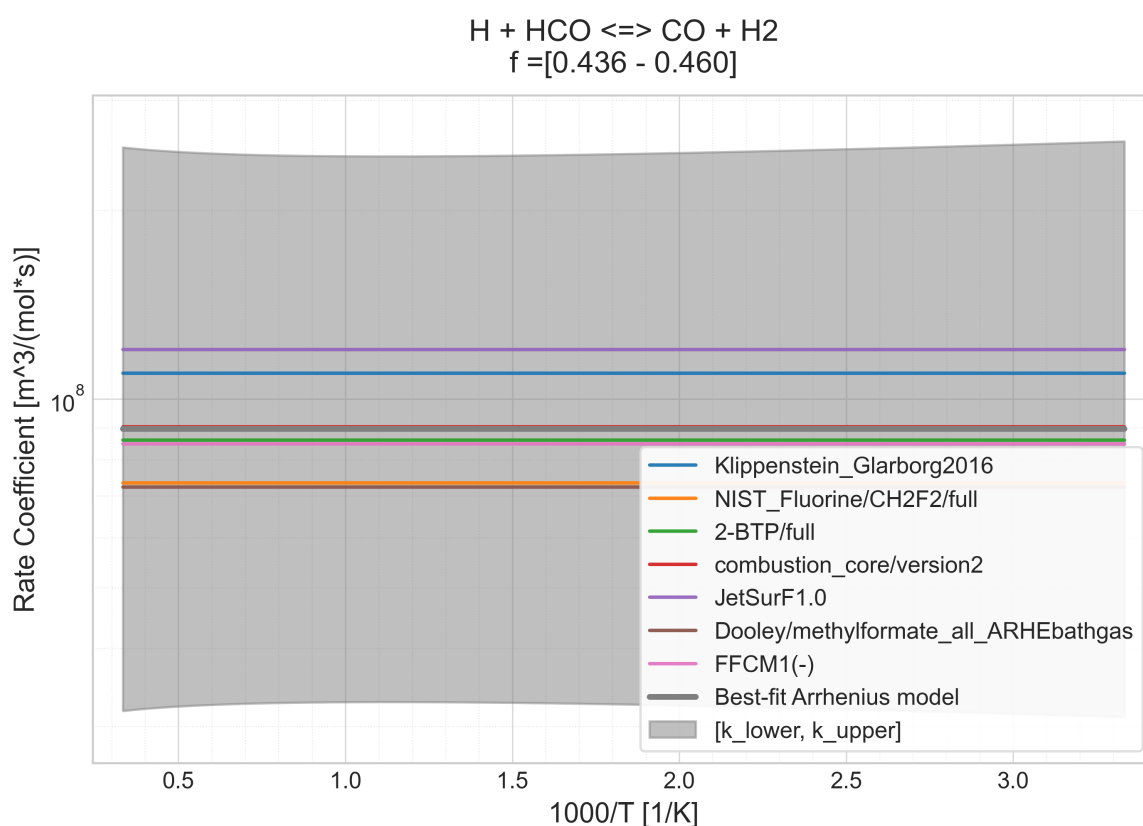


Figure 4.11: Uncertainty factor for $H + HCO \rightleftharpoons CO + H_2$ [6]

Uncertainty factors have been calculated using all available libraries in RMG database. If there are not enough sources in the database for a given reaction, uncertainty factor calculation cannot be performed.

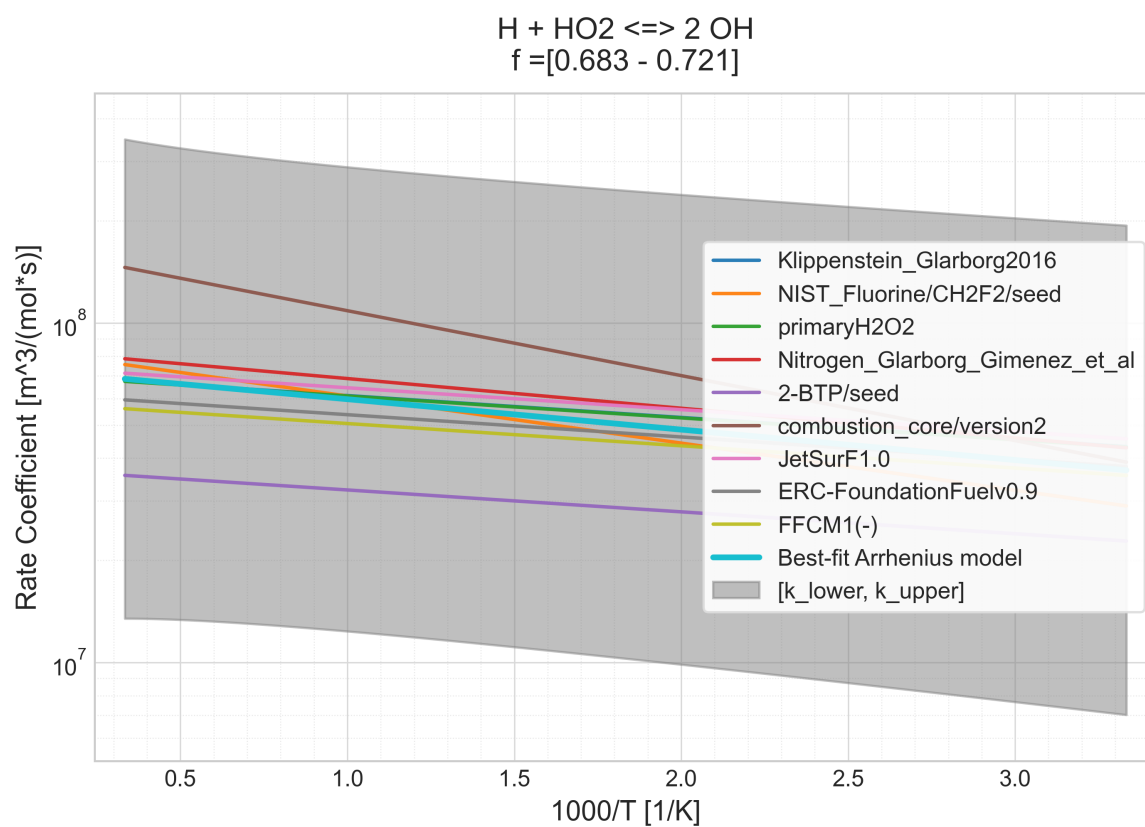


Figure 4.12: Uncertainty factor for $H + HO_2 \rightleftharpoons 2 OH$ [6]

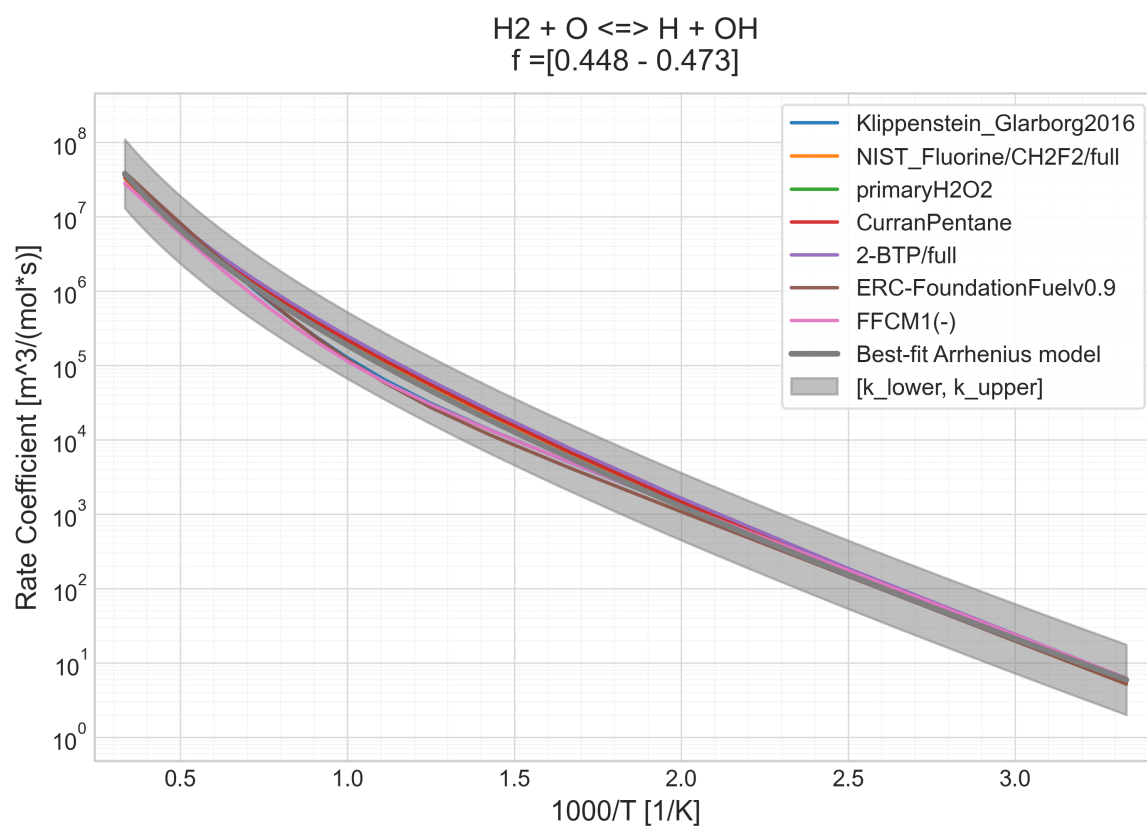


Figure 4.13: Uncertainty factor for $H_2 + O \rightleftharpoons H + OH$ [6]

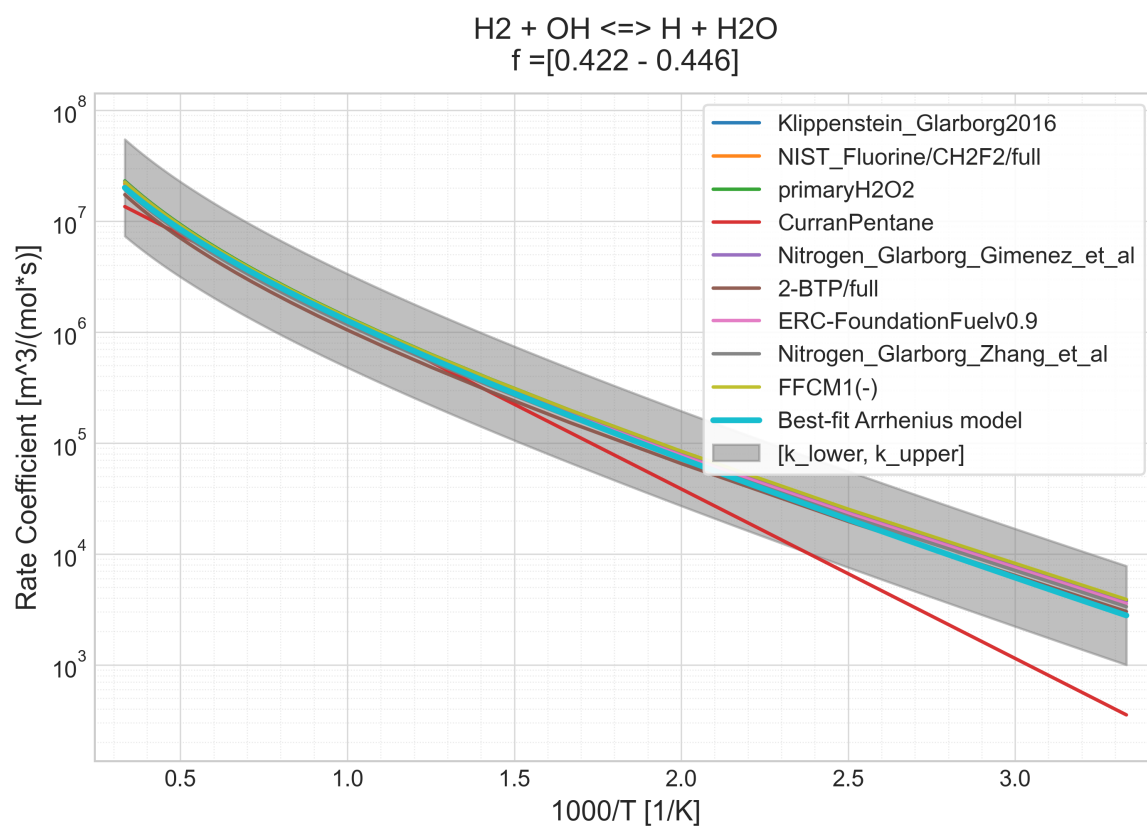


Figure 4.14: Uncertainty factor for $\text{H}_2 + \text{OH} \rightleftharpoons \text{H} + \text{H}_2\text{O}$ [6]

4.4 Cantera Shock Tube IDT Simulations

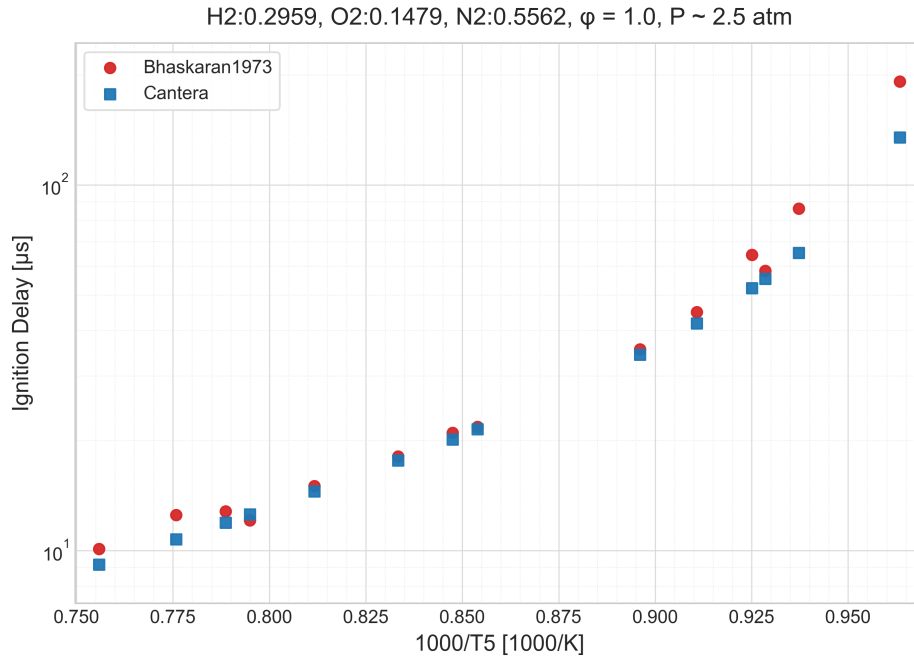


Figure 4.15: Comparison of Cantera IDT simulation with experimental measurement [49] IDT Type: dP/dt max.

Current implementation of Cantera IDT simulation is tested with all available Syngas and Hydrogen IDT data from RESPETCH database. Figure 4.15, Figure 4.16, Figure 4.17 show some example comparisons from that dataset.

Figure 4.17 illustrates the discrepancy between constant volume reactor Cantera simulation and the real shock tube experiment, where boundary layer buildup reduces the effective volume. This pressure rise should be compensated for in simulation to get more accurate IDT simulations. However in the RESPETCH syngas/hydrogen IDT datasets, pressure rise is rarely reported. It would be valuable for TUMKIN database to be more comprehensive.

4.4.0.1 Reaction Mechanisms

Reaction mechanisms used to validate the software developed in this thesis are Ethylene [51] and Combined Hydrogen Syngas [52].

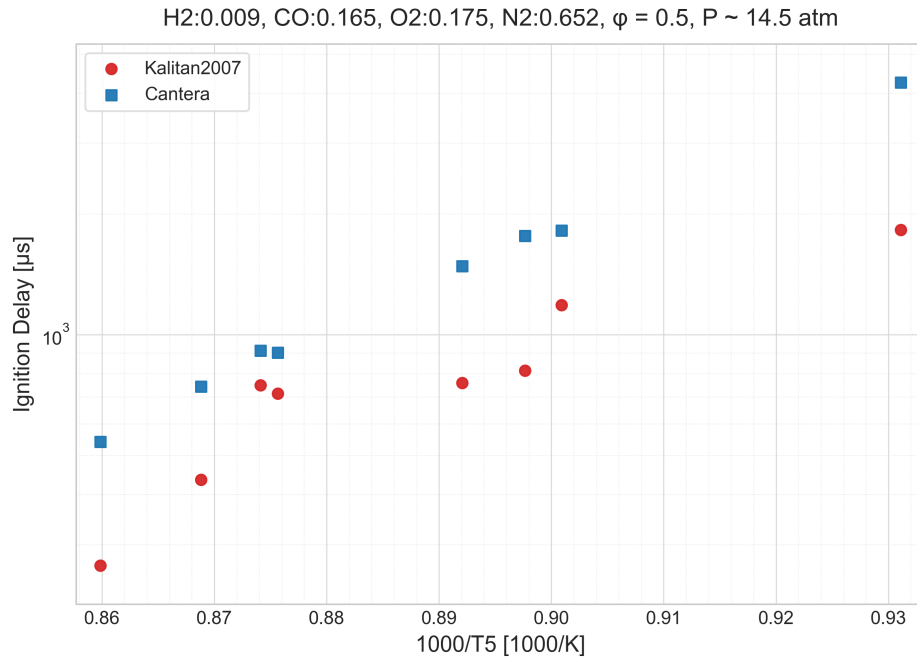


Figure 4.16: Comparison of Cantera IDT simulation with experimental measurement [46] IDT Type: $d(OH^*)/dt$ max.

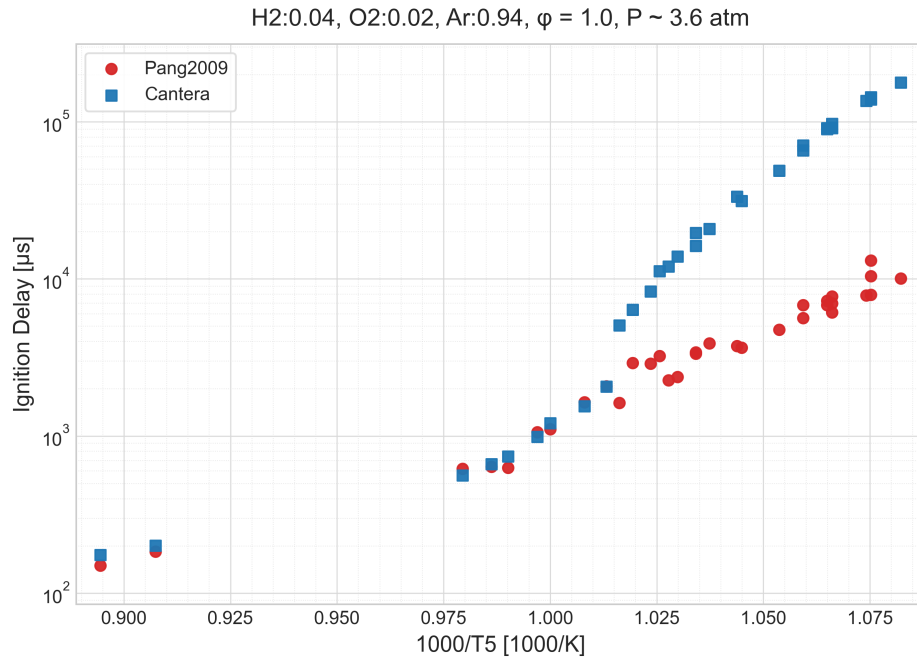


Figure 4.17: Comparison of Cantera IDT simulation with experimental measurement [50] IDT Type: OH max.

5 Conclusion and Outlook

The primary objective of this thesis was to develop and implement a robust framework for evaluating data consistency in reaction model optimization. To this end, a reaction rate coefficient uncertainty package was developed to estimate model uncertainties, and an ignition delay time uncertainty package was implemented to quantify experimental data uncertainties. These components were integrated into the Bound-to-Bound Data Collaboration (B2BDC) framework to perform comprehensive consistency analyses. To support those functionalities, Cantera ignition delay time simulations were implemented. Sensitivity analysis was implemented to determine active reactions and reduce computational cost of the rest of the analysis. The results of these analyses are heavily dependent on the quality and availability of the underlying data. Although literature sources for ignition delay time often report critical parameters such as T_5 , P_5 , ϕ uncertainties, or pressure rise, this information is rarely integrated into large experimental databases. A similar challenge exists regarding kinetics databases. While the NIST Chemical Kinetics Database [39] is highly comprehensive, it cannot be accessed programmatically. Conversely, the RMG database [53] is accessible and user-friendly but often lacks essential information, such as specific temperature ranges for kinetic data. Future work should focus on the following areas:

- Full integration of the packages developed in this thesis into the TUMKIN infrastructure to streamline the optimization workflow.
- Building a comprehensive database for TUMKIN that supports the analysis and optimization of a wider range of reaction models and fuels.
- Extending the uncertainty quantification methods used for ignition delay times to other experimental targets, such as laminar flame speeds and species concentrations.
- Exploring the implementation of model discrepancy functions to explicitly account for inherent model inadequacies both during consistency analysis and the optimization process.

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Declaration

Affirmation

Hereby, I affirm that I am the sole author of this thesis. To the best of my knowledge, I affirm that this thesis does not infringe upon anyone's copyright nor violate any proprietary rights. I affirm that any ideas, techniques, quotations, or any other material, are in accordance with standard referencing practices.

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Munich, March 2, 2026

BARKIN GÜNEY